MADUGUNDU VISHNU VARDHAN

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PROFILE

Detail-oriented technology professional with a background in Electrical and Electronics Engineering and hands-on experience in robotics, embedded systems, programming, and technical instruction. Skilled in delivering coding and robotics workshops, guiding learners through practical projects, and integrating hardware and software technologies. Strong in problem-solving, team collaboration, technical communication, and project execution, with a focus on leading technical initiatives, mentoring teams, and developing innovative technology solutions.

EDUCATION

G Pullaiah College of Engineering and Technology **July 2024**
B.Tech in Electrical and Electronics Engineering CGPA: 8.97

A P Model School and Junior College **April 2020**
Intermediate CGPA: 7.90

Z P High School **March 2018**
SSC CGPA: 9.30

SKILLS SUMMARY

- **Programming & Development** : Python, Arduino Programming, JavaScript, HTML, CSS
- **Data Base** : MySQL
- **Computer Vision & IoT** : OpenCV (cv2), Hand Gesture Recognition, ESP32, Serial Communication
- **Tools/SDLC** : VS Code, Windows, Agile(sdlc)
- **Soft Skills** : Critical-Thinking, Problem Solving, Communication Teaching, Adaptability.

WORK EXPERIENCE

Robotics Instructor at Nxtwave:(working here)

- Designed and delivered robotics training programs for undergraduate students, covering Python programming, ROS, robotics fundamentals, and IoT technologies.
- Conducted hands-on robotics and IOT workshops to help students assemble, program, and test robots using Arduino, esp32, sensors, etc..
- Guided students in developing robotics and IoT-based projects using Python and ROS, focusing on practical applications and problem-solving.
- Managed robotics lab equipment, troubleshot hardware and software issues, and collaborated with faculty and trainers to improve course content.

Coding and Robotics/STEM Instructor at Wizklub:(2+ years)

- Delivered coding education to students for multiple grades, adapting teaching methods based on learning levels.
- Introduced programming concepts using interactive tools like Blockly and Python, web designing and embedded C.
- Designed engaging lesson plans incorporating games, challenges, and real-world applications.
- Improved student participation and understanding through hands-on activities and project-based learning.

PROJECTS

Design of Novel Multilevel Inverter for Hybrid Renewable Energy Sources

- **Technologies Used :** MLI, H-Bridge Inverters, OPAL-RT, PV-System.
- Developed a new circuit with a constant number of switches and achieved a pure sinusoidal waveform. My specific contributions included designing the circuit architecture and optimizing the switching sequence to reduce harmonic distortion.

AI Gesture Controlled LED System

- **Technologies:** Python, OpenCV (cv2), Arduino/ESP32, Serial Communication
- Developed a real-time gesture-controlled system to change LED colors using computer vision. Utilized OpenCV for hand gesture recognition and integrated it with ESP32 through serial communication. Implemented hand landmark detection and distance-based gesture logic to enable dynamic color selection (Red, Green, Blue, White) and LED ON/OFF control, ensuring smooth interaction between software and hardware.

CERTIFICATES

- Appreciation certificate by HCL Jigsaw for training the children for the competition.
- Certified in python by Infosys Spring Board.
- Achieved publication certificate for the project publication.